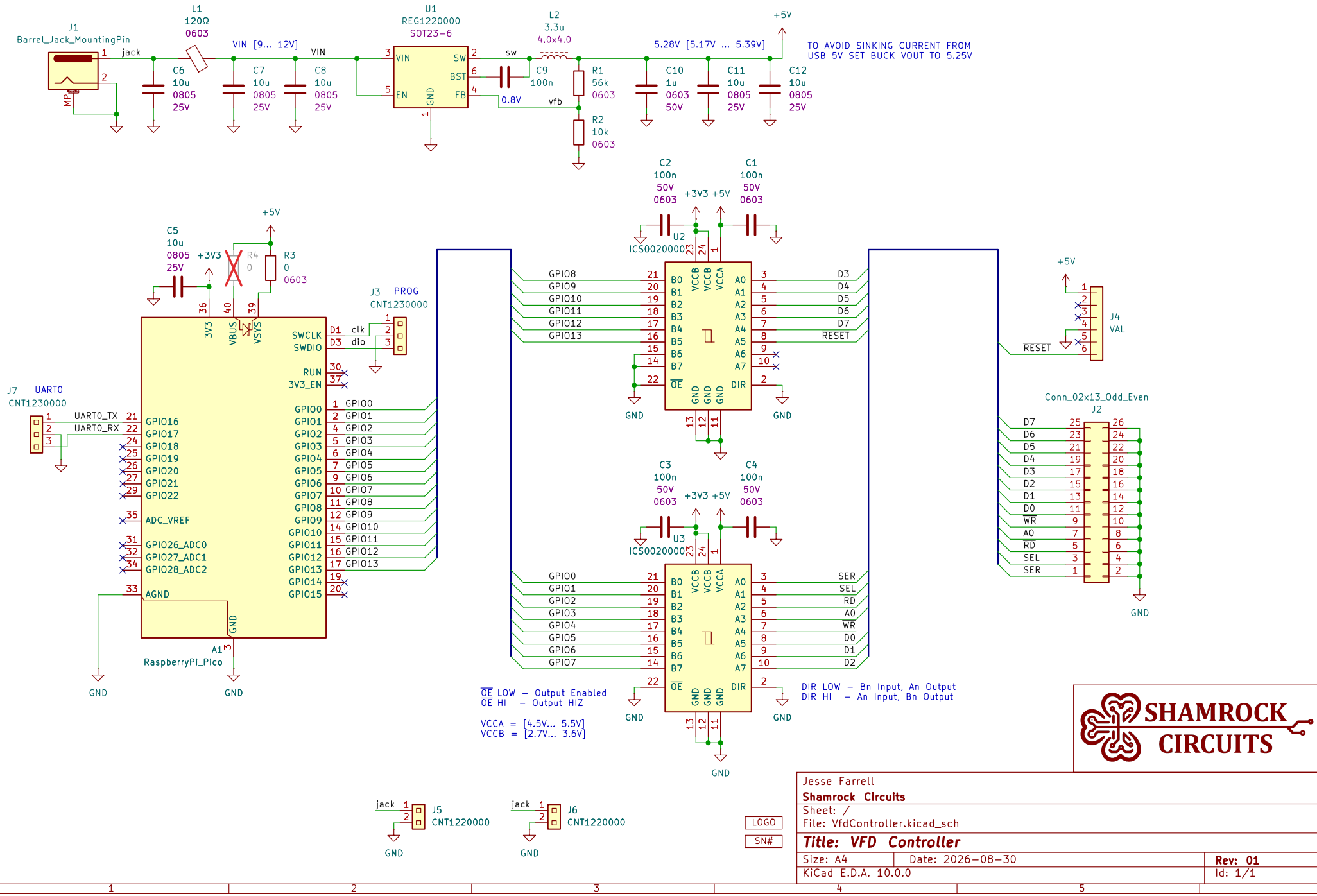

J1
Barrel_Jack_MountingPin
L1
120Ω
0603
U1
REG1220000
SOT23-6
L2
3.3u
4.0x4.0
+5V
5.28V [5.17V ... 5.39V]
TO AVOID SINKING CURRENT FROM USB 5V SET BUCK VOUT TO 5.25V
VIN [9... 12V]
C6 10u 0805 25V
C7 10u 0805 25V
C8 10u 0805 25V
C9 100n
R1 56k 0603
R2 10k 0603
C10 1u 0603 50V
C11 10u 0805 25V
C12 10u 0805 25V
0.8V
vfb
C2 100n 50V 0603
C1 100n 50V 0603
U2
ICS00200000
C3 100n 50V 0603
C4 100n 50V 0603
U3
ICS00200000
C5 10u 0805 25V
R3 0 0603
R4 0
J3 PROG
CNT1230000
J7 UART0
CNT1230000
UART0_TX
UART0_RX
RaspberryPi_Pico
J4 VAL
Conn_02x13_Odd_Even
J2
OE LOW – Output Enabled
OE HI – Output HIZ
VCCA = [4.5V... 5.5V]
VCCB = [2.7V... 3.6V]
DIR LOW – Bn Input, An Output
DIR HI – An Input, Bn Output
J5 CNT1220000
J6 CNT1220000
LOGO
SN#
SHAMROCK CIRCUITS
Jesse Farrell
Shamrock Circuits
Sheet: /
File: VfdController.kicad_sch
Title: VFD Controller
Size: A4
Date: 2026-08-30
Rev: 01
KiCad E.D.A. 10.0.0
Id: 1/1